

GPT-2 Arm-ו Retrieval ,HUB_DIM Collapse

סיכום והסבר ברור של הדיון: בחירת ממד ה-Hub, שחזור, Verification, R@1, SVD, AUC, anisotropy, hubness וקריסת זרוע GPT-2

GPT-2

Verification

Retrieval

Structural Similarity

Shared Hub

הרעיון המרכזי: HUB_DIM צריך להיות גדול מספיק כדי לשמר מידע, אך לא גדול עד כדי כך שהמיפוי אליו נהיה חלש סטטיסטית. במקביל, ביצועי retrieval יכולים לקרוס גם כאשר עדיין נשמר signal חלש יותר שנראה ב-Verification AUC.

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- 1. בחירת HUB_DIM על סמך מדידה
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- 10. סיכום קצר לבחינה / דו"ח

1. בחירת HUB_DIM על סמך מדידה

HUB_DIM הוא מספר הממדים של המרחב המשותף - ה-Hub - שאליו ממפים ייצוגים ממודלים או מרחבים שונים.



הבחירה ב-HUB_DIM היא trade-off בין שתי דרישות מנוגדות:

בחירה	יתרון	סיכון
Hub צר	קל יותר ללמוד mapping אמין	compression חזק מדי ואובדן מידע
Hub רחב	יכולת טובה יותר לשמר/לשחזר מידע	פחות דוגמאות לכל ממד; mapping יכול להפוך underpowered

לכן לא בוחרים את הערך לפי אינטואיציה בלבד. מבצעים **sweep** על מספר ערכים ומודדים reconstruction/retention, יחס rows-per-dimension ותוצאת transfer.

המסר של המשפט "the choice is made from evidence rather than argued":
לא אומרים בדיעבד "1024 נשמע סביר". מראים ניסויית איזה ערך מספק את האיזון הטוב ביותר.

2. למה Hub קטן מדי מאבד מידע

נניח שמרחב היעד הוא ברוחב 2048 ממדים, אך ה-Hub הוא רק 256 ממדים:

$$2048 \rightarrow 256 \rightarrow 2048$$

אם המבנה החשוב של היעד דורש יותר מ-256 כיוונים עצמאיים, ה-Hub הופך ל-**bottleneck**. הוא אינו יכול לשמר את כל המידע.

$$B \rightarrow H \rightarrow B_{\text{hat}}$$

If the hub is too narrow:

$$B_{\text{hat}} \neq B$$

זו המשמעות של:

“the hub cannot RECONSTRUCT a wide target”

כלומר, הבעיה אינה רק "ממד קטן" באופן מופשט; היא נמדדת בפועל באמצעות איכות reconstruction או retention.

3. למה Hub גדול מדי עלול להיות underpowered

ככל שמספר ממדי ה-Hub גדל, יש יותר דרגות חופש במיפוי שצריך ללמוד. מספר הדוגמאות, לעומת זאת, נשאר קבוע.

אם יש 10,000 rows:

$$\text{HUB_DIM} = 256$$

$$\text{rows per dimension} = 10000 / 256 \approx 39.1$$

$$\text{HUB_DIM} = 2048$$

$$\text{rows per dimension} = 10000 / 2048 \approx 4.88$$

בפרויקט שעליו דיברנו, הסף שנקבע הוא לפחות:

$$N / \text{HUB_DIM} \geq 5$$

כאשר היחס יורד מתחת לסף, המיפוי אל ה-Hub מקבל פחות תמיכה סטטיסטית לכל ממד. במילים פשוטות: מנסים לאמוד מרחב רחב מדי ביחס לכמות הנתונים.

the largest useful dimension היא ה-Hub הרצוי אינו בהכרח הרחב ביותר, אלא **Sweet spot**:
supported by the data

4. SVD slicing ו- Sweep

במקום לבחור ערך בודד, בודקים למשל:

256, 512, 768, 1024, 1536, 2048

ולכל רוחב מודדים לפחות:

1. **Reconstruction / retention** - כמה מהמידע המקורי נשמר.
2. **Rows per hub dimension** - האם יש מספיק נתונים ביחס לרוחב.
3. **Transfer performance** - האם ההתאמה בין המרחבים נשמרת לאחר המעבר דרך ה-Hub.

אם משתמשים ב-SVD:

$$X = U \Sigma V^T$$

מחשבים את ה-SVD פעם אחת, ואז משתמשים ב-prefix שונה של הרכיבים:

Hub 256 → first 256 singular components
Hub 512 → first 512 singular components
Hub 1024 → first 1024 singular components

לכן ניתן לבצע sweep על מספר רוחבים בעלות נוספת קטנה יחסית.

5. R@1 לעומת AUC Verification

R@1 - שאלת retrieval חזקה

עבור query מסוים A_i , יש הרבה candidates בצד השני:

$$B_1, B_2, \dots, B_n$$

וה-partner האמיתי הוא B_i . R@1 שואל:

$$\operatorname{argmax}_j \operatorname{sim}(A_i, B_j) = i ?$$

כלומר: האם הזוג הנכון הוא **מקום ראשון מכל המועמדים**?

Verification AUC - שאלה חלשה יותר

כאן שואלים האם score של true pair נוטה להיות גבוה מ-score של random pair:

$$\text{sim}(A_i, B_i) > \text{sim}(A_i, B_j), \quad j \neq i$$

או באופן הסתברותי:

$$P(\text{score_true} > \text{score_random})$$

AUC גבוה פירושו שיש יכולת טובה להפריד בין positive pair ל-negative/random pair, אך הוא אינו דורש שה-positive יהיה המועמד מספר 1 מתוך אלפים.

דוגמה: AUC גבוה אך R@1 נמוך

נניח שה-zוג האמיתי מקבל:

$$\text{true pair} = 0.80$$

ורוב הזוגות הלא נכונים מקבלים 0.30-0.60, אך קיימים שני negatives עם:

$$0.83, 0.82$$

ה-zוג האמיתי עדיין נראה הרבה יותר טוב מזוג אקראי טיפוסי, ולכן AUC Verification יכול להיות גבוה. אבל הוא אינו במקום הראשון, ולכן R@1 עבור אותה query נכשל.

מסקנה: Verification בודק אם נשאר correspondence signal. Retrieval, ובמיוחד R@1, בודק אם ה-signal חד מספיק כדי לזהות את ה-partner הנכון מבין הרבה מועמדים.

6. מהו GPT-2 arm collapse?

בהקשר של ניסוי shared-hub / structural similarity, הביטוי מתאר מצב שבו **הצד של GPT-2 מאבד discrimination** גיאומטרי שימושי.

Model A $\longrightarrow$ HUB $\longleftarrow$ GPT-2

במצב תקין, ייצוגי GPT-2 שונים אמורים לשמר הבדלים משמעותיים:

$g_1, g_2, g_3, \dots$ are meaningfully distinct

ב-collapse, הרבה ייצוגים נעשים דומים מדי מבחינה גיאומטרית:

$g_1 \approx g_2 \approx g_3 \approx \dots$

התוצאה: ה-score של ה-target הנכון אינו מובחן מספיק מה-scores של candidates שגויים.

למה cosine גבוה אינו בהכרח טוב?

לפני collapse אפשר לראות:

true similarity = 0.85
random similarity = 0.20

אחרי collapse:

true similarity = 0.92
random similarity = 0.88

למרות שה-cosine של הזוג הנכון עלה, ה-margin בין true ל-random כמעט נעלם. לכן הייצוג פחות שימושי ל-retrieval.

7. הקשר בין collapse, anisotropy ו-hubness

Anisotropy

במרחב isotropic, vectors מפוזרים בהרבה כיוונים. ב-anisotropy חזק, רבים מהם מצביעים לכיוון כללי דומה או נמצאים בתוך cone צר.

Isotropic: ↗ ← ↓ ↘ ↑
 Anisotropic: ↗ ↗ ↗ ↗ ↗

כאשר קיים common component חזק, cosine similarity יכול להיות גבוה גם לזוגות שאינם תואמים.

Hubness

Hubness הוא מצב אחר: candidates מסוימים מופיעים כ-nearest neighbor של מספר חריג של queries.

many queries → the same few candidate vectors

לכן:

מושב	מה הוא מתאר?
Anisotropy	הווקטורים מרוכזים מדי בכיוונים דומים.
Hubness	ווקטורים מסוימים נהפכים לשכנים של יותר מדי queries.
Arm collapse	התוצאה הפרקטית: הצד של GPT-2 אינו מספק מספיק discrimination ל-transfer/retrieval-אמין.

8. למה AUC יכול לשרוד כאשר R@1 קורס?

נניח שאחרי transfer מתקבלים scores:

True pairs: 0.87, 0.89, 0.86, 0.90
 Random pairs: 0.75, 0.78, 0.80, 0.77

ברמת distribution, true pairs עדיין נוטים להיות גבוהים יותר, ולכן AUC נשאר טוב.

אבל retrieval מתבצע מול אלפים או עשרות אלפים של candidates. מספיק שמספר קטן של negatives יקבלו scores גבוהים במיוחד כדי לעקוף את הזוג הנכון:

wrong candidates: 0.91, 0.90, 0.89

לכן ייתכן:

Verification AUC = good
 R@1 = poor

הפרשנות: נשאר signal גם של correspondence, אך אבד חלק מהמידע המקומי/הפרטני הדרוש ל-exact matching.

9. Intrinsic GPT-2 anisotropy לועות hub-induced collapse

חשוב להבדיל בין שתי תופעות:

א. אנאיזוטרופיה פנימית של GPT-2

הבעיה קיימת כבר בייצוג המקורי של GPT-2 לפני שמיפינו אותו ל-Hub. אם representations מקוריים כבר קרובים מאוד זה לזה, ה-Hub מתחיל מנקודת פתיחה קשה.

ב. Collapse שנגרם על ידי ה-Hub

כאן אובדן ההבחנה נוצר או מחמיר כתוצאה מ-mapping, low-rank truncation, projection, לא יציב או bottleneck קטן מדי.

GPT-2: 768 dims → Hub: 64 dims → reconstructed GPT-2

אם הכיוונים שמבדילים בין examples נמצאים ברכיבים שנחתכו, אז לאחר reconstruction מתקבלים vectors דומים יותר:

original: g_1, g_2, g_3 are distinct
reconstructed: $\hat{g}_1 \approx \hat{g}_2 \approx \hat{g}_3$

שרשרת סיבתית אפשרית

GPT-2 anisotropy
↓
less effective variation
↓
restrictive hub / projection
↓
additional loss of discrimination
↓
GPT-2 arm collapse
↓
R@1 crashes
↓
AUC may still remain respectable

10. מה זה אומר לגבי Structural Similarity?

Collapse בצד GPT-2 אינו מוכיח בהכרח שאין shared structure בין המודלים.

ייתכנו לפחות שתי פרשנויות:

1. אין מספיק structural correspondence אמיתי.
2. יש correspondence, אך measurement pipeline איבד את היכולת לפתור אותו ברזולוציה הנדרשת.

אם AUC עדיין גבוה בזמן ש-R@1 קרס, האפשרות השנייה נהיית סבירה יותר: יש signal שנשמר, אבל אינו מספיק ל-identification חד-חד-ערכי.

ניסוח שימושי לדו"ח:

The GPT-2 arm may preserve coarse correspondence while losing pair-specific discriminability. A collapse in R@1 therefore does not by itself establish absence of shared structure; it may instead indicate that the mapping or hub geometry no longer resolves that structure at retrieval precision

11. סיכום קצר לבחינה / דו"ח

- HUB_DIM קטן מדי → reconstruction/retention → bottleneck נפגעים.
- HUB_DIM גדול מדי → פחות rows לכל mapping → dimension עלול להיות underpowered.
- בוחרים Hub באמצעות sweep מדיד, לא באמצעות נימוק אינטואיטיבי בלבד.
- SVD ניתן לחשב פעם אחת ואז לחתוך לרוחבים שונים.
- R@1 שואל האם ה-target האמיתי הוא מקום ראשון מתוך כל candidates.
- Verification AUC שואל האם true pairs מובחנים סטטיסטית מ-random pairs.
- לכן AUC יכול להישאר גבוה גם כאשר R@1 נמוך.
- GPT-2 arm collapse = אובדן discriminative geometry בצד GPT-2.
- Anisotropy = כיוונים דומים מדי; hubness = מסוימים נהפכים לשכנים של יותר מדי queries.
- Collapse אינו לבדו הוכחה להעדר structural similarity; הוא יכול להצביע על כשל רזולוציה של ה-pipeline.

המסמך נערך מתוך הדיון הנוכחי בצ'אט, בניסוח לימודי ומאורגן.

Update: G1/G2 Shared Hub, Head, Query Flow, and New-Model Integration

Added from the recent discussion. This section separates the Hub, encoder-to-Hub maps, the Hub-to-bge head, E1's direct DINOv2-to-bge experiment, query-time retrieval, and GPT-2's role.

Most important distinction: the 512-d object belongs to G1's shared Hub. It is not the same thing as E1's direct DINOv2-to-bge map. In G1 the path is encoder -> Hub(512) -> head -> bge(1024).

10. Predicted effective rank

Predicted effective rank means the number of directions expected to carry useful, independent signal, even when the nominal vector width is much larger.

nominal dimension = 768, but usable effective rank may be about 6.5

If most spectral mass sits in only a few singular directions, the space behaves as if it were much lower dimensional. This matters because a wide space can still have very poor cosine discrimination if nearly all vectors occupy the same narrow cone.

GPT-2 example from the project: raw GPT-2 target geometry had effective rank about 6.5/768 and mean pair-cosine about +0.999. That is why raw cosine can look almost perfect while retrieval remains poor.

11. What are the 8,533 "same items"?

G1 uses 9,533 aligned semantic items in total: **8,533 for fitting** and **1,000 held out for evaluation**.

"Same item" does **not** mean every model receives the same raw input. It means the rows refer to the same underlying COCO image-caption item:

Branch	Input for row i	Output width
DINOv2-small	image i (pixels)	768
DINOv2-base	image i (pixels)	1536
DINOv2-large	image i (pixels)	2048
bge-m3	caption(s) describing image i	1024
GPT-2	caption(s) describing image i	768
BERT	caption(s) describing image i	768
SBERT	caption(s) describing image i	768

Example row:

Image: a cat sitting on a chair.

Text: "A cat is sitting on a chair."

The three DINO branches encode the pixels. bge-m3, GPT-2, BERT, and SBERT encode text describing the same referent.

12. How the 512-d Hub is created

For each of the 8,533 fitting items, take the seven aligned representations and standardize each model space separately.

$$768 + 1536 + 2048 + 1024 + 768 + 768 + 768 = 7680$$

Then concatenate the seven standardized representations row by row:

$$z_i = [x_{\text{small}} \mid x_{\text{base}} \mid x_{\text{large}} \mid x_{\text{bge}} \mid x_{\text{gpt2}} \mid x_{\text{bert}} \mid x_{\text{sbert}}] \text{ in } \mathbb{R}^{7680}$$

Across all fitting rows:

$$Z \text{ in } \mathbb{R}^{(8533 \times 7680)}$$

Apply PCA/SVD, keep the top 512 components, then whiten them. The resulting training-coordinate matrix is:

$$H \text{ in } \mathbb{R}^{(8533 \times 512)}$$

What is stored? The project can cache the 8,533 training Hub vectors, but the important reusable objects are the Hub basis/normalization and the model-specific linear maps. For a new item, its 512-d Hub vector is computed on the fly.

13. One encoder-to-Hub map per model

Once H is defined, each encoder learns how to express its own vectors in the common 512-d coordinates:

$$X_k W_k \approx H$$

Map	Shape
W_{small}	768 x 512
W_{base}	1536 x 512
W_{large}	2048 x 512
W_{bge}	1024 x 512
W_{gpt2}	768 x 512
W_{bert}	768 x 512
W_{sbert}	768 x 512

At runtime only one encoder is needed. Example:

new image -> DINOv2-base -> $x_{\text{base}}(1536)$ -> W_{base} -> $h(512)$

The original 6144-d concatenation is a **Hub-construction operation**, not a production inference requirement.

14. How the caption-retrieval head is created

The head is a separate linear map:

$$W_{\text{head}} : \mathbb{R}^{512} \rightarrow \mathbb{R}^{1024}$$

It is fitted using only the DINOv2-small arm after that arm has already entered the Hub:

$$H_{\text{small}} W_{\text{head}} \approx T_{\text{bge}}$$

Here H_{small} contains 512-d Hub vectors produced from DINOv2-small, and T_{bge} contains the corresponding native bge-m3 caption targets.

Why this is a strong test: after fitting, W_{head} is frozen. The same head is then fed Hub vectors coming from DINOv2-base and DINOv2-large, which it never saw during head fitting. The reported transfer reaches 95.9% and 92.9% of the corresponding native-head performance.

15. What does t -hat mean?

The hat means **predicted**.

$$t_{\text{hat}_i} = h_i W_{\text{head}}$$

t_i would be the native bge-m3 embedding of the caption. t_{hat_i} is the image's predicted vector in bge-m3 space after going through the Hub and head.

image -> DINO -> Hub vector h_i -> W_{head} -> predicted bge vector t_{hat_i}

16. Query flow example: "cat sit on a chair"

Offline image indexing

image_j -> DINOv2-base -> W_{base} -> $h_j(512)$ -> W_{head} -> $t_{\text{hat}_j}(1024)$

Store each t_{hat_j} in the retrieval index.

Online text query

$q = \text{bge}(\text{"cat sit on a chair"})$ in $\mathbb{R}^{1024}$

Now both query and indexed images are in bge space:

$\text{score}_j = \text{cosine}(q, t_{\text{hat}_j})$

Rank the gallery by score and return the top-k images.

The Hub is the interoperability layer; bge-m3 is the final retrieval space.

The normal text query does not need to pass through GPT-2 or through the Hub because bge already produces a native vector in the target retrieval space.

17. How GPT-2 can contribute

1. **During Hub construction:** GPT-2 caption representations are one of the seven representation blocks used to define the joint PCA/whitened Hub basis.

2. **As its own Hub arm:** GPT-2 has a fitted map $W_{\text{gpt2}} : 768 \rightarrow 512$, so GPT-2 text can in principle be placed into the Hub.
3. **Not required for normal bge retrieval:** for a text query, $\text{text} \rightarrow \text{bge}$ is the direct and appropriate path.

A possible experimental alternative is:

$\text{text} \rightarrow \text{GPT-2} \rightarrow W_{\text{gpt2}} \rightarrow \text{Hub} \rightarrow W_{\text{head}} \rightarrow \text{bge space}$

But the supplied project results do not establish this exact end-to-end path as a high-performing deployed query route. GPT-2's raw geometry is especially problematic because its effective rank is very low and cosine is nearly degenerate.

18. Adding a new encoder to the existing Hub

Suppose an eighth, new vision model outputs $d_{\text{new}} = 1280$ dimensions.

Preferred modular integration

1. Keep the existing 512-d Hub fixed.
2. Run the new model on aligned items for which the desired Hub coordinates are already known.
3. Fit only one new ridge map:

$$X_{\text{new}} W_{\text{new}} \approx H, \quad W_{\text{new}} \in \mathbb{R}^{(1280 \times 512)}$$

Then use:

$\text{new model} \rightarrow W_{\text{new}} \rightarrow \text{existing Hub} \rightarrow \text{existing } W_{\text{head}} \rightarrow \text{bge space}$

Engineering advantage: adding a model can mean fitting one new model-to-Hub map rather than building pairwise adapters to every existing model.

If you rebuild the Hub

If you concatenate six spaces and rerun PCA, the Hub axes can change. Even if the width remains 512, the old head may no longer interpret the new coordinates correctly. A rebuilt Hub can therefore require refitting or realigning existing maps and heads.

This frozen-Hub eighth-model integration is the natural architectural extension of G1, but the supplied report does not claim it has already been demonstrated for an arbitrary unseen sixth encoder.

19. Do not confuse E1 with G1

	E1 direct cross-modal map	G1 shared-Hub system
Core path	DINOv2(768) -> W -> bge(1024)	encoder -> Hub(512) -> head -> bge(1024)
Training data	8,000 COCO train2017 image-caption pairs	9,533 aligned items total; 8,533 fit + 1,000 eval
Fit/eval	7,000 fit + 1,000 held out	8,533 fit + 1,000 held out
Caption treatment	5 human captions encoded and averaged for one target	Text branches participate in the joint Hub and task-head evaluation
512-d object?	No shared 512-d Hub in the direct E1 map	Yes - central shared Hub

E1's five-caption averaging improved held-out R-squared from 0.278 to 0.511. The direct fit used $7,000/768 = 9.1$ rows per input dimension, with 1,000 rows held out. The matched ceiling must be measured on those same 1,000 rows, in the same image-to-text direction and with the same averaging protocol; under the matched protocol, the reported result is 40.9% of ceiling rather than the earlier inflated 55.2%.

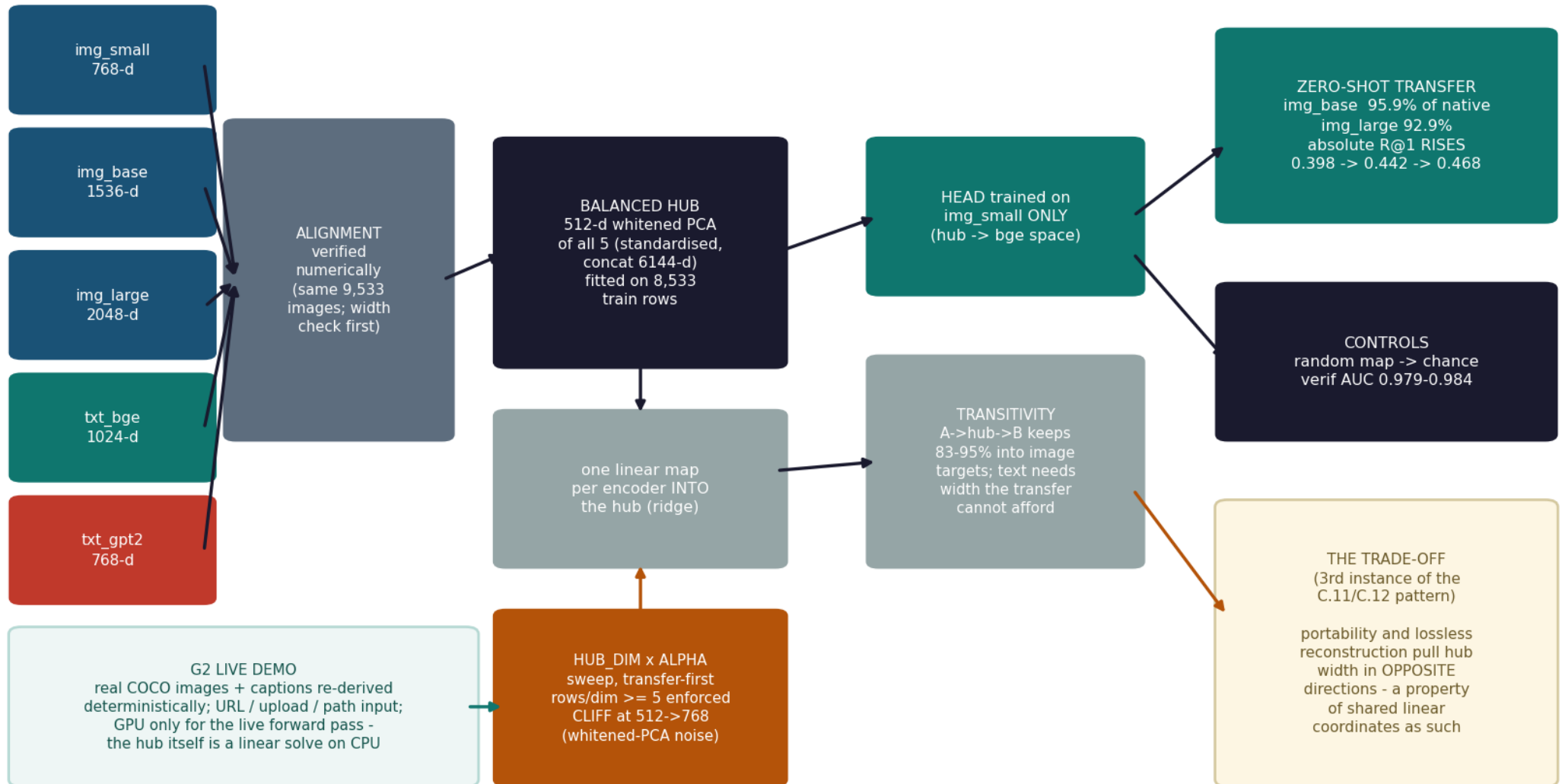
20. One-line mental model

G1: build a common 512-d language from seven representation spaces; teach each encoder one linear map into that language; train one task head on one encoder's Hub vectors; test whether the same head works when another encoder speaks the same Hub language.

encoder -> W_encoder -> HUB_512 -> W_head -> BGE_1024 -> cosine retrieval

Visual A - Original G1/G2 project flow

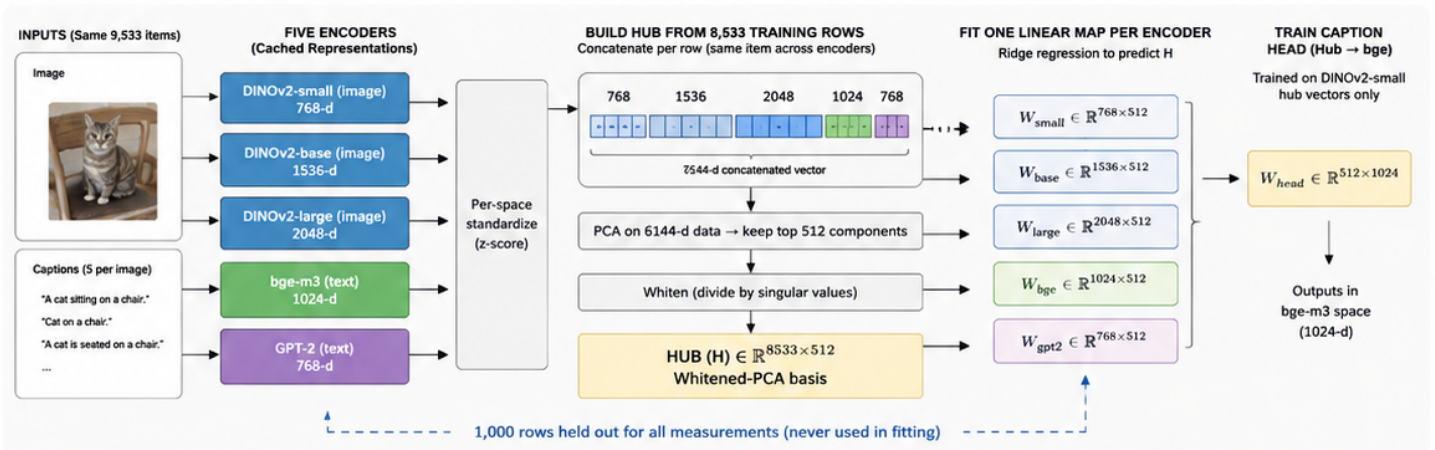
G1/G2 — five encoders, one 512-d hub: the shared space built, measured, and demonstrated live



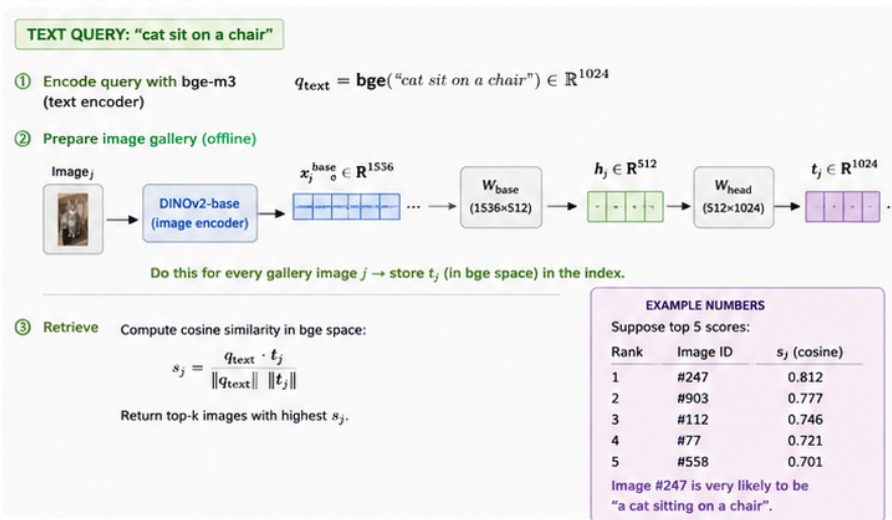
Historical five-space G1/G2 figure retained for context. The current G1 run has seven encoders; updated measured figures follow in the 7-space appendix.

Visual B - Early full-system infographic

1. SYSTEM OVERVIEW (G1: Five Encoders → One 512-d Hub)



2. QUERY FLOW EXAMPLE



3. MATH SUMMARY

Hub construction

- Standardize each space: $\tilde{X}_k = \frac{X_k - \mu_k}{\sigma_k}$
- Concatenate per row: $Z = [\tilde{X}_{\text{small}} \mid \tilde{X}_{\text{base}} \mid \tilde{X}_{\text{large}} \mid \tilde{X}_{\text{bge}} \mid \tilde{X}_{\text{gpt2}}] \in \mathbb{R}^{8533 \times 7544}$
- PCA (SVD): $Z = U \Sigma V^T$
- Keep top 512: $V_{512} \in \mathbb{R}^{6144 \times 512}$
- Whiten: $H = Z V_{512} \Sigma_{512}^{-1} \in \mathbb{R}^{8533 \times 512}$

Encoder → Hub maps (ridge regression)

For encoder k with matrix $X_k \in \mathbb{R}^{8533 \times d_k}$:

$$W_k = \arg \min_W \|X_k W - H\|_F^2 + \alpha \|W\|_F^2$$

Closed form: $W_k = (X_k^T X_k + \alpha I)^{-1} X_k^T H \in \mathbb{R}^{d_k \times 512}$

Caption head (Hub → bge)

Using only small encoder hub vectors $H_{\text{small}} \in \mathbb{R}^{8533 \times 512}$ and bge targets $T_{\text{bge}} \in \mathbb{R}^{8533 \times 1024}$:

$$W_{\text{head}} = \arg \min_W \|H_{\text{small}} W - T_{\text{bge}}\|_F^2 + \lambda \|W\|_F^2$$

$$W_{\text{head}} = (H_{\text{small}}^T H_{\text{small}} + \lambda I)^{-1} H_{\text{small}}^T T_{\text{bge}} \in \mathbb{R}^{512 \times 1024}$$

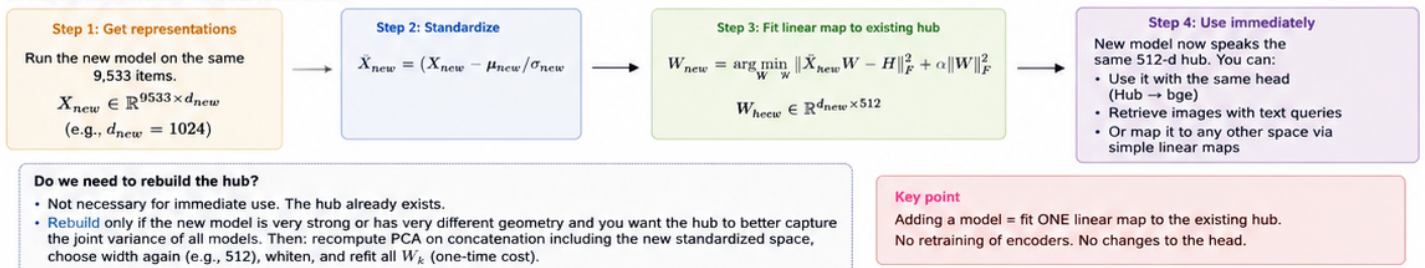
4. DETAILED DIMENSIONS

Component	Dim In	Dim Out	Shape
DINOv2-small	768	512	$W_{\text{small}} \in \mathbb{R}^{768 \times 512}$
DINOv2-base	1536	512	$W_{\text{base}} \in \mathbb{R}^{1536 \times 512}$
DINOv2-large	2048	512	$W_{\text{large}} \in \mathbb{R}^{2048 \times 512}$
bge-m3	1024	512	$W_{\text{bge}} \in \mathbb{R}^{1024 \times 512}$
GPT-2	768	512	$W_{\text{gpt2}} \in \mathbb{R}^{768 \times 512}$
Hub → bge head	512	1024	$W_{\text{head}} \in \mathbb{R}^{512 \times 1024}$

WHY IT WORKS

- Hub is learned from all five spaces together (whitened PCA).
- Each encoder learns how to express itself in that common coordinate system (linear maps).
- Head learned on one encoder's hub vectors can be reused on other encoders (zero-shot component transfer).
- Retrieval happens in bge space, where text queries live natively.

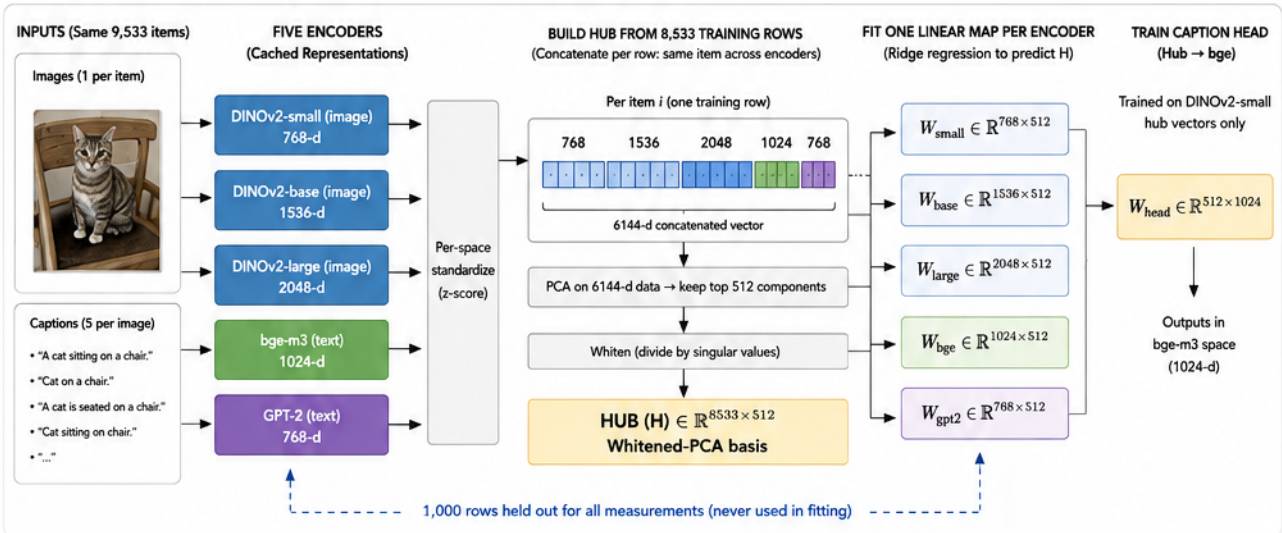
5. ADDING A NEW MODEL (Scenario)



Earlier five-space visual retained for conceptual context; the updated seven-space measurements are shown later.

Visual C - Higher-resolution version with legend

1. SYSTEM OVERVIEW (G1: Five Encoders → One 512-d Hub,



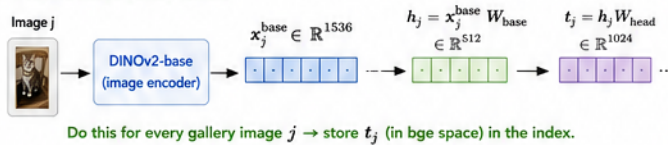
2. QUERY FLOW EXAMPLE

TEXT QUERY: "cat sit on a chair"

- ① Encode query with bge-m3 (text encoder)

$$q_{\text{text}} = \text{bge}(\text{"cat sit on a chair"}) \in \mathbb{R}^{1024}$$

- ② Prepare image gallery (offline)



- ③ Retrieve (online)

Compute cosine similarity in bge space:

$$s_j = \frac{\langle q_{\text{text}}, t_j \rangle}{\|q_{\text{text}}\| \|t_j\|}$$

Return top-k images with highest s_j .

EXAMPLE SCORES (top 5)

Rank	Image ID	s_j (cosine)
1	#247	0.812
2	#903	0.777
3	#112	0.746
4	#77	0.721
5	#558	0.701

Image #247 is very likely to be "a cat sitting on a chair".

3. MATH SUMMARY

Hub construction

- Standardize each space: $\tilde{X}_k = \frac{X_k - \mu_k}{\sigma_k}$
- Concatenate per row: $Z = [\tilde{X}_{\text{small}} | \tilde{X}_{\text{base}} | \tilde{X}_{\text{large}} | \tilde{X}_{\text{bge}} | \tilde{X}_{\text{gpt2}}] \in \mathbb{R}^{8533 \times 6144}$
- PCA (SVD): $Z = U \Sigma V^T$
- Keep top 512: $V_{512} \in \mathbb{R}^{6144 \times 512}$
- Whiten: $H = Z V_{512} \Sigma_{512}^{-1} \in \mathbb{R}^{8533 \times 512}$

Encoder → Hub maps (ridge regression)

For encoder k with matrix $X_k \in \mathbb{R}^{8533 \times d_k}$:

$$W_k = \arg \min_W \|X_k W - H\|_F^2 + \alpha \|W\|_F^2$$

Closed form: $W_k = (X_k^T X_k + \alpha I)^{-1} X_k^T H \in \mathbb{R}^{d_k \times 512}$

Caption head (Hub → bge)

Using only small encoder hub vectors $H_{\text{small}} \in \mathbb{R}^{8533 \times 512}$ and bge targets $T_{\text{bge}} \in \mathbb{R}^{8533 \times 1024}$:

$$W_{\text{head}} = \arg \min_W \|H_{\text{small}} W - T_{\text{bge}}\|_F^2 + \lambda \|W\|_F^2$$

Closed form: $W_{\text{head}} = (H_{\text{small}}^T H_{\text{small}} + \lambda I)^{-1} H_{\text{small}}^T T_{\text{bge}} \in \mathbb{R}^{512 \times 1024}$

4. DETAILS

DIMENSIONS TABLE

Component	Dim In	Dim Out	Matrix Shape
DINOv2-small	768	512	$W_{\text{small}} \in \mathbb{R}^{768 \times 512}$
DINOv2-base	1536	512	$W_{\text{base}} \in \mathbb{R}^{1536 \times 512}$
DINOv2-large	2048	512	$W_{\text{large}} \in \mathbb{R}^{2048 \times 512}$
bge-m3	1024	512	$W_{\text{bge}} \in \mathbb{R}^{1024 \times 512}$
GPT-2	768	512	$W_{\text{gpt2}} \in \mathbb{R}^{768 \times 512}$
Hub → bge head	512	1024	$W_{\text{head}} \in \mathbb{R}^{512 \times 1024}$

WHY IT WORKS

- ✓ Hub is learned from all five spaces together (whitened PCA).
- ✓ Each encoder learns how to express itself in that common coordinate system (linear maps).
- ✓ Head learned on one encoder's hub vectors can be reused on other encoders (zero-shot component transfer).
- ✓ Retrieval happens in bge space, where text queries live natively.

LEGEND

- Image encoders (DINOv2 family)
- Text encoder (contrastive)
- Text encoder (language model)
- Learned objects (Hub, Head)
- Preprocessing / operations
- Hub space (512-d)
- bge-m3 space (1024-d)

KEY NOTATION

- X_k : raw matrix of encoder k
- $\tilde{X}_k$: standardized (z-score)
- Z : concatenated matrix (6144-d)
- H : hub matrix (512-d)
- W_k : encoder → hub map
- W_{head} : hub → bge head
- q_{text} : text query vector in bge space (1024-d)
- t_j : image j after head in bge space (1024-d)
- $\langle \cdot, \cdot \rangle$: dot product
- $\|\cdot\|$: L2 norm
- $R@K$: Recall@K (retrieval metric)

5. ADDING A NEW MODEL (Scenario)

Step 1: Get representations

Run the new model on the same 9,533 items (or more).

$$X_{\text{new}} \in \mathbb{R}^{9533 \times d_{\text{new}}}$$

(e.g., $d_{\text{new}} = 1024$)

Step 2: Standardize

$$\tilde{X}_{\text{new}} = \frac{X_{\text{new}} - \mu_{\text{new}}}{\sigma_{\text{new}}}$$

Step 3: Fit linear map to existing hub

$$W_{\text{new}} = \arg \min_W \|\tilde{X}_{\text{new}} W - H\|_F^2 + \alpha \|W\|_F^2$$

Closed form:

$$W_{\text{new}} = (\tilde{X}_{\text{new}}^T \tilde{X}_{\text{new}} + \alpha I)^{-1} \tilde{X}_{\text{new}}^T H$$

$$W_{\text{new}} \in \mathbb{R}^{d_{\text{new}} \times 512}$$

Step 4: Use immediately

New model now speaks the same 512-d hub. You can:

- Use it with the same head (Hub → bge)
- Retrieve images with text queries
- Or map it to any other space via simple linear maps

Do we need to rebuild the hub?

- Not necessary for immediate use. The hub already exists.
- Rebuild only if the new model is very strong or has very different geometry and you want the hub to better capture the joint variance of all models. Then: recompute PCA on concatenation including the new standardized space, choose width again (e.g., 512), whiten, and refit all W_k (one-time cost).

Key point

Adding a model = fit ONE linear map to the existing hub. No retraining of encoders. No changes to the head.

IMPORTANT TAKEAWAYS



One hub for many encoders:
All models map linearly into the same 512-d space.



Component transfer:
A caption head trained on one encoder works on others it never saw (zero-shot).



Retrieval in text space:
Everything is converted to bge-m3 space (1024-d) for searching with text queries.



Extensible by design:
New models plug in by fitting one linear map—no retraining, no changes to the head.

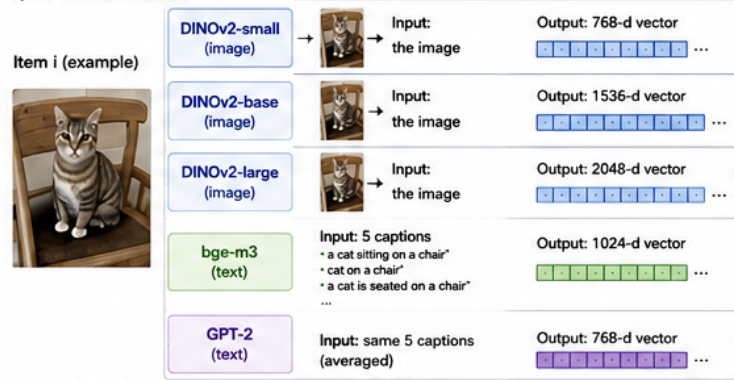
Earlier five-space high-resolution visual with legend; retained for conceptual context.

Visual D - Simple model-input and GPT-2 contribution view

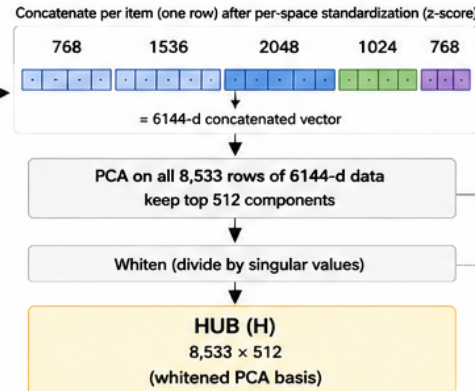
PART A. SYSTEM OVERVIEW (How the hub and maps are built)

We start with the SAME 8,533 training items (images with captions).

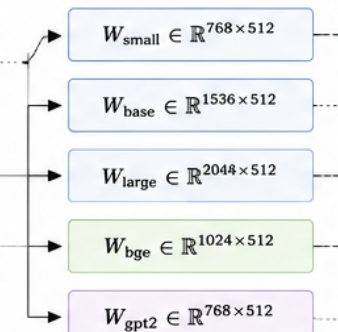
1) INPUT PER MODEL



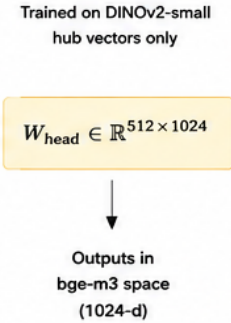
2) BUILD THE HUB (from all models together)



3) FIT ONE LINEAR MAP PER MODEL (Ridge regression to predict H)



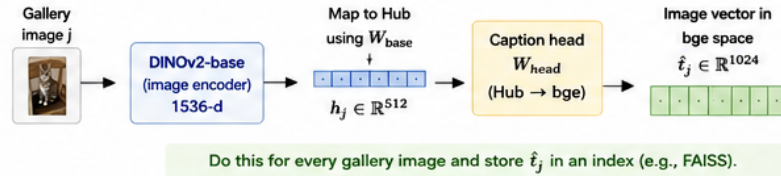
4) TRAIN CAPTION HEAD (Hub → bge)



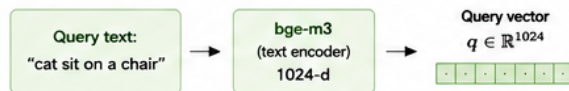
1,000 rows held out for all measurements (never used in fitting)

PART B. QUERY FLOW EXAMPLE (Find images for: "cat sit on a chair")

1) Prepare image gallery (offline once)



2) Process text query (online)



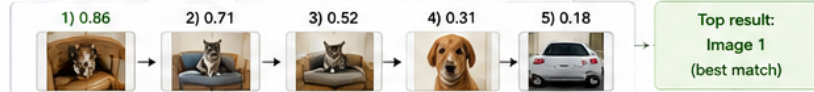
3) Retrieve

Compute cosine similarity in bge space

$$s_j = \frac{\langle q, \hat{t}_j \rangle}{\|q\| \|\hat{t}_j\|}$$

Return top-k images with highest s_j

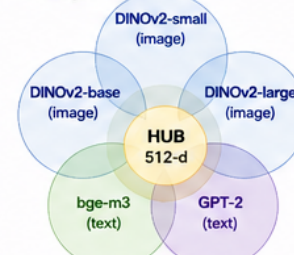
Example similarity scores (top 5)



PART C. CONTRIBUTION OF GPT-2 (How and if it helps)

During training (hub construction)

GPT-2 text representations are one of the five spaces used to build the hub. They influence the hub basis so that it captures directions useful for both images and language, even from a language model that was never trained with images.



At query time (does GPT-2 run?)

✓ NORMAL PATH (recommended)
Use bge-m3 to encode the query text. (Faster and much better retrieval quality.)
Text (query) → bge-m3 → compare with image vectors (in bge space).

? ALTERNATIVE PATH (possible but weaker)
You could use GPT-2 to encode the query and map it to hub:
Text → GPT-2 → W_{gpt2} → Hub → W_{head} → bge space → compare
This works, but typically lower accuracy than using bge-m3.

So GPT-2's main contribution is in building a better shared hub, not in being the best encoder for queries.

LEGEND

- Image encoder (DINOv2 family)
- Text encoder (contrastive: bge-m3)
- Text encoder (language model: GPT-2)
- Learned component (Hub / Head)
- Operations / processing
- bge-m3 space (1024-d)
- Hub space (512-d)

KEY DIMENSIONS

DINOv2-small	768 → 512
DINOv2-base	1536 → 512
DINOv2-large	2048 → 512
bge-m3	1024 → 512
GPT-2	768 → 512
Hub → bge head	512 → 1024

KEY TAKEAWAYS

- ★ One shared 512-d hub is learned from all five models together (8,533 training items).
- ★ Each model gets one linear map to the hub (ridge regression).
- ★ A caption head maps hub vectors to bge-m3 space (1024-d).
- ★ For text queries, use bge-m3 (best). GPT-2 can be used alternatively via Hub → Head, but is weaker.
- ★ The head trained on DINOv2-small transfers to unseen encoders (base/large) at ~93–96% of native performance.
- ★ 1,000 items are held out for all measurements (never used in fitting).

Earlier five-space simplified view of per-model inputs, Hub construction, query flow, and GPT-2's optional role.

Visual E - Latest direct DINOv2-to-bge explanation (Version 1: full)

1. THE DATA (SUPERVISED BY HUMAN ANNOTATION)

IMAGE-CAPTION PAIRS (COCO)

Row i is an image AND the captions describing it.
Different objects sharing a referent.

Row i example
(one of 8,000)



5 human captions
(COCO gives five per image)

- 1 "A cat sitting on a chair.
- 2 "Cat on chair.
- 3 "A cat is seated on a chair.
- 4 "Cat sitting on the chair.
- 5 "A cat perched on a wooden chair."
- ...

5 CAPTIONS AVERAGED

Each caption is encoded with bge-m3. The five embeddings are averaged into one target vector.

→ Makes the target "what this image is about" rather than "what one annotator wrote".

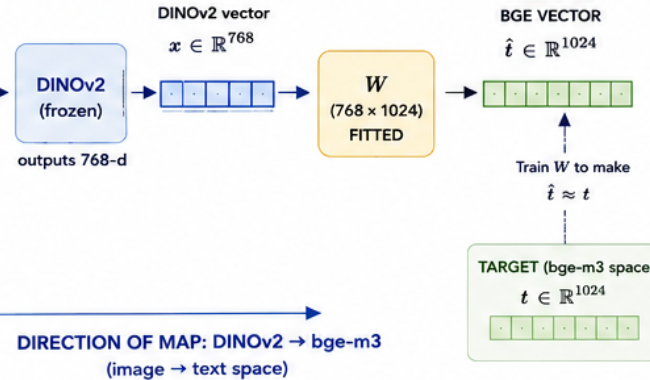


8k COCO train2017

- 8,000 such pairs drawn from COCO's training split (118k images).
- Not val2017 (5,000 images) because we need enough rows left after the eval holdout.

2. OVERALL FLOW (DINOv2 → BGE-M3)

Images go into DINOv2, one matrix W converts the resulting vectors into bge-m3's text space. Both models are frozen; W is the only thing fitted.



3. HOW W IS FITTED (RIDGE REGRESSION)

Fit 7,000 (9.1 rows/dim)

- 7,000 rows train the ridge.
- DINOv2 outputs 768 numbers.
- $\frac{7,000}{768} = 9.1$ rows per input dimension.
- Project rule: ≥ 5 required, 5–10 working band.
- Experiment A successful run sat at 9.8.
→ 9.1 is comfortably powered.
- Checked BEFORE results are read, so a weak outcome can't be blamed on starvation.

Ridge regression:

$$W = \arg \min_W \|XW - T\|_F^2 + \alpha \|W\|_F^2$$

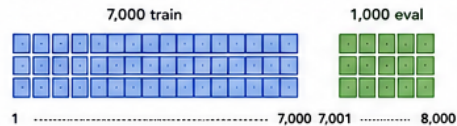
$X \in \mathbb{R}^{n \times 768}$ (DINOv2 vectors)

$T \in \mathbb{R}^{n \times 1024}$ (bge-m3 targets)

$n = 7,000$ (training rows)

4. TRAIN / EVAL SPLIT

Total: 8,000 pairs (COCO train2017)



TRAIN (7,000)

Used to fit W .
 W only sees these rows.



EVAL (1,000)

Held out completely.
Never used in fitting.

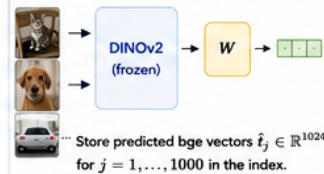


W is then frozen and used for evaluation only.

5. QUERY / RETRIEVAL FLOW (IMAGE → TEXT)

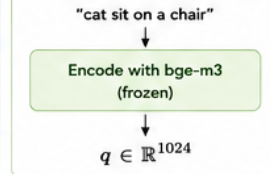
1) Encode eval gallery (offline)

Each of the 1,000 eval images:



2) Process caption query (online)

Query caption (averaged like training):



3) Retrieve (image → text)

For $j = 1 \dots 1000$, compute cosine similarity in bge space:

$$s_j = \frac{\langle q, \hat{t}_j \rangle}{\|q\| \|\hat{t}_j\|}$$

Return top- k images with highest s_j .

Example top-5 scores ($k=5$)



Each caption-vector queries a gallery of 1,000 candidates → chance = $1 / 1000 = 0.001$.

6. CEILING MEASUREMENT (CRITICAL FOR % SCORE)

Ceiling measured on the SAME 1,000 rows

- This is the clause that took three attempts today.
- The result is reported as a % of what a JOINTLY TRAINED model achieves, because captions are ambiguous and no independent pair can beat a model built for the task.
- Ceiling must be measured on the SAME rows, in the same direction (image→text), with the same caption treatment (averaged). When it wasn't, the score read 55.2%; matched properly, 40.9%.

Example (on the same 1,000 eval rows)

System	R@1 (image→text)	
Our linear map (DINOv2 → bge-m3)	0.409	(40.9% of ceiling)
Jointly trained model (ceiling)	1.000	(100%)
Chance (1/1000)	0.001	(0.1%)



THE WHOLE LINE IN ONE SENTENCE: 8,000 human-paired image-caption rows, 7,000 to fit one matrix at comfortable statistical power, 1,000 held out, and the benchmark computed on those same 1,000 rows under identical conditions so the percentage means.

LEGEND

- Image encoder (DINOv2)
- Text encoder (GPT-2)
- Learned component (Matrix W or Head)
- Operations / processing
- Text encoder (bge-m3)

$x \in \mathbb{R}^{768}$

$t \in \mathbb{R}^{1024}$

$\hat{t} \in \mathbb{R}^{1024}$

DINOv2 vector (image)

bge-m3 target vector (text)

Predicted bge vector

$W \in \mathbb{R}^{768 \times 1024}$

$q \in \mathbb{R}^{1024}$

s_j

R@K

Fitted linear map (only trained thing)

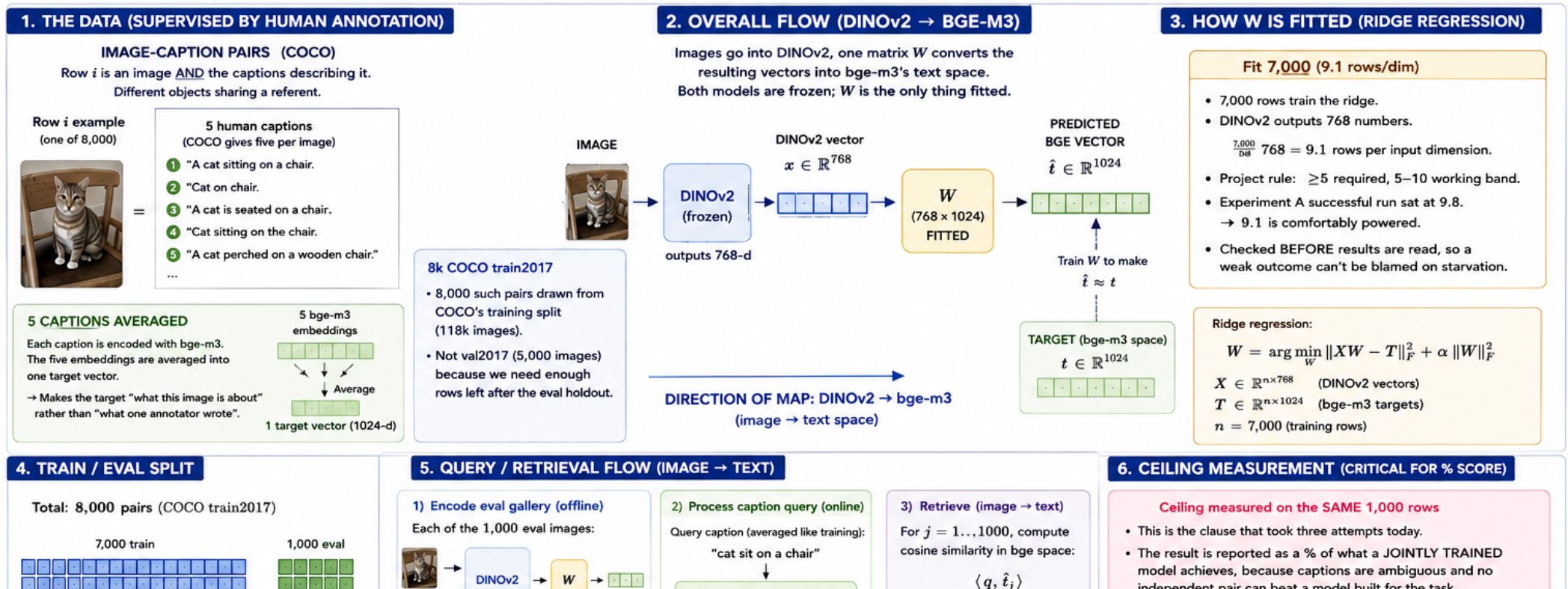
Query vector (text)

Cosine similarity score

Recall@K

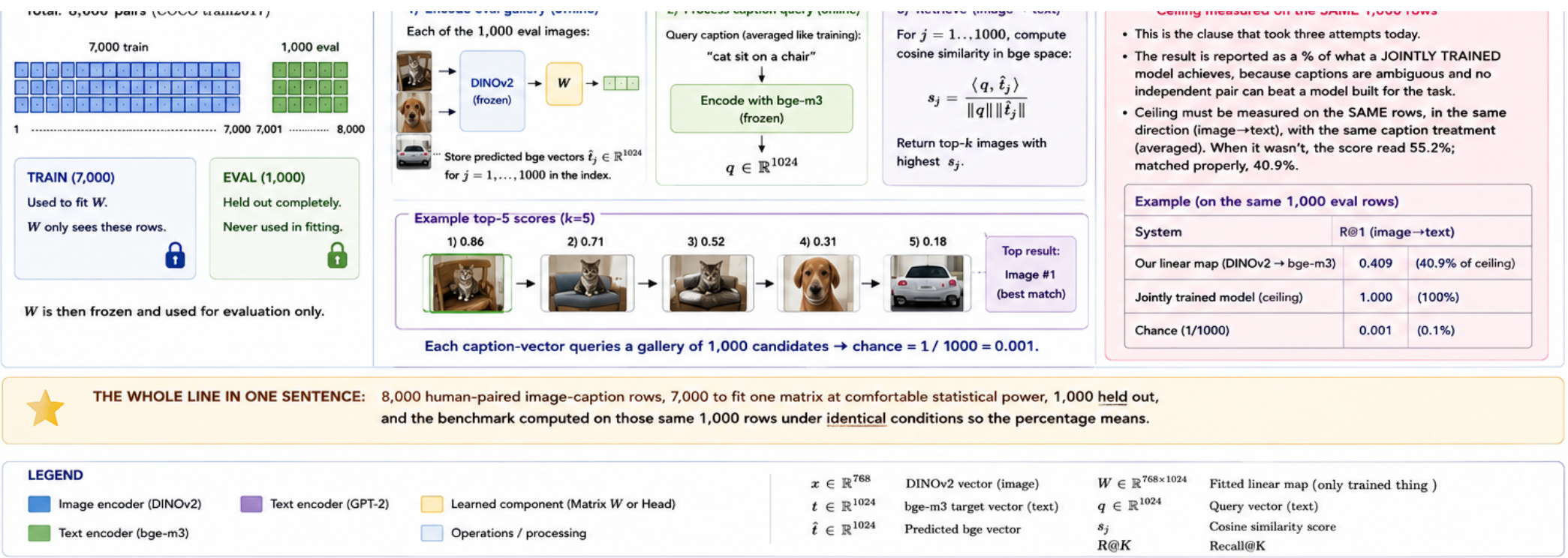
Full latest version: image-caption pairs, 5-caption averaging, 7,000 fit / 1,000 eval, direct DINOv2 → bge map, and matched ceiling.

Visual E - Latest direct DINOv2-to-bge explanation (Version 2: enlarged top)



Reader-friendly enlarged view of the top portion: data, map direction, and ridge fit.

Visual E - Version 2 continued: enlarged bottom



Reader-friendly enlarged view of the bottom portion: train/eval split, query flow, ceiling, and legend.

2026-08-16 Update: G1 Seven-Space Hub - Tests, Geometry, Portability, and Transfer

This appendix supersedes the earlier five-space G1 descriptions where they conflict. The current run uses seven representation spaces on 9,533 common aligned items.

Current G1 scope: DINOv2-small, DINOv2-base, DINOv2-large, bge-m3, GPT-2, BERT, and SBERT. The Hub is a balanced 512-d whitened-PCA coordinate system. Each encoder has its own linear map into the Hub.

21. Data inventory, common-item intersection, and fitting power

Space	Cached rows x width	Rows used in common G1 set	Train rows / input dim
img_small	9,533 x 768	9,533	11.1
img_base	18,066 x 1,536	9,533	5.6
img_large	23,755 x 2,048	9,533	4.2
txt_bge	18,066 x 1,024	9,533	8.3
txt_gpt2	18,066 x 768	9,533	11.1
txt_bert	18,066 x 768	9,533	11.1
txt_sbert	18,066 x 768	9,533	11.1

The common aligned set contains **9,533 items**: 8,533 train and 1,000 evaluation. Shape agreement uses 400 held-out items and fits no map.

Power nuance: the project's Hub-width sweep enforces adequate rows per *Hub dimension*; however, rows per encoder input dimension are not identical. In particular, `img_large` has $8,533/2,048 = 4.2$ rows per input dimension. That should be reported explicitly rather than described as comfortably above a 5-per-input-dimension rule.

22. Test 1 - Raw shape agreement before the Hub exists

Question: without fitting any map, do two spaces already agree on the relative geometry of the same held-out items?

Method: for each encoder pair, compute pairwise cosine distances among the same 400 held-out items, then correlate the two distance rankings with Spearman rho. A high rho means the two spaces preserve a similar near/far ordering even though their coordinates, dimensions, and training objectives may differ.

Interpretation: this is a geometry test, not a coordinate-alignment test. High rho means a correspondence is already partly present; the Hub only has to supply coordinates for it.

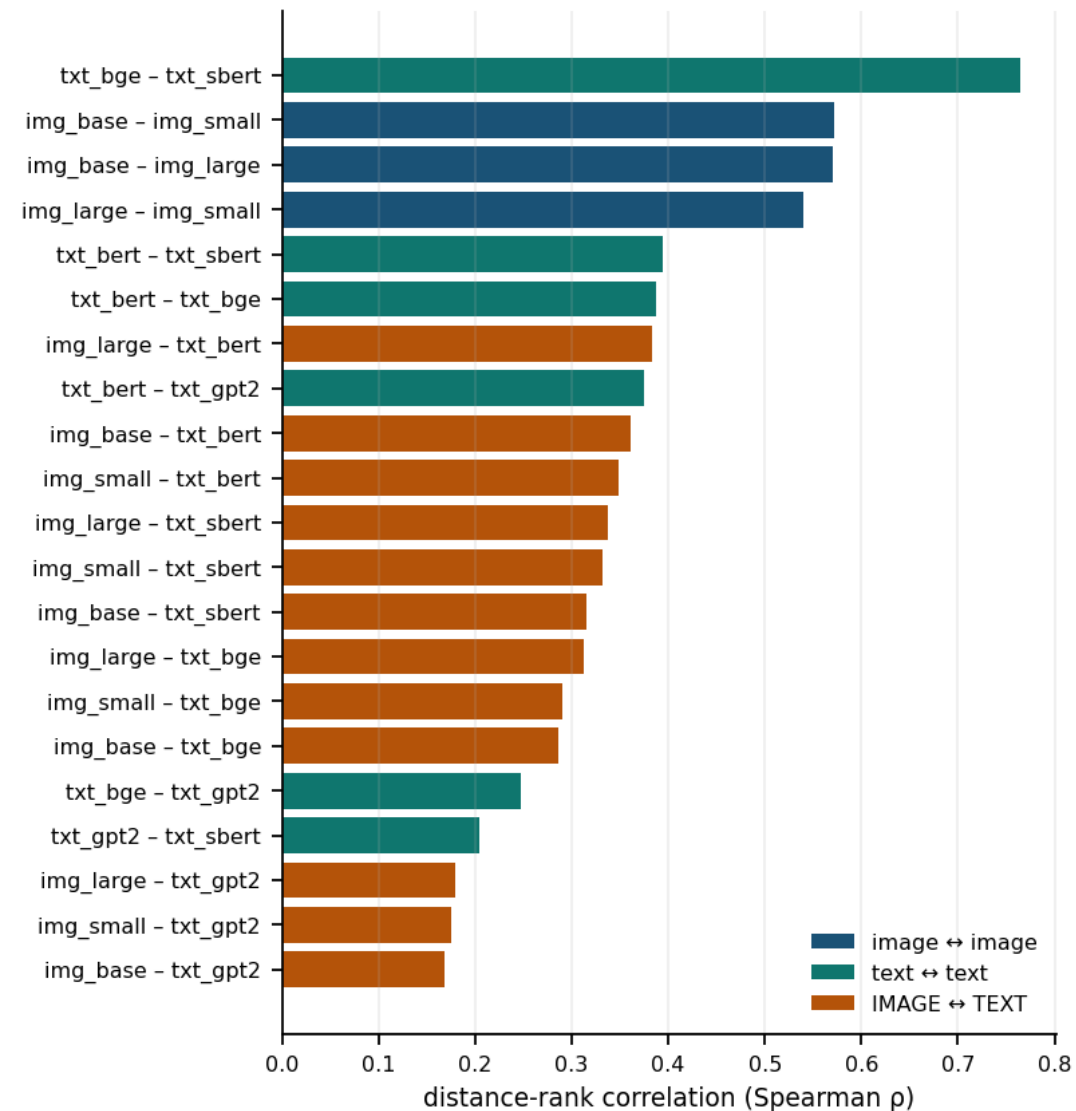
Group	Mean rho	Meaning
image-image	+0.561	Strongest group, but DINOv2 small/base/large share ancestry through the DINOv2 family.
image-text	+0.291	Cross-modal structure is real before fitting; image and text encoders agree modestly on the relative arrangement of the same semantic items.
text-text	+0.395	Substantial within-text agreement across different objectives.

The strongest pair overall is `txt_bge <-> txt_sbert` at $\rho = +0.764$. That exceeds the image-image group mean, which is important because bge and SBERT share a retrieval/similarity-oriented objective but not DINOv2 lineage.

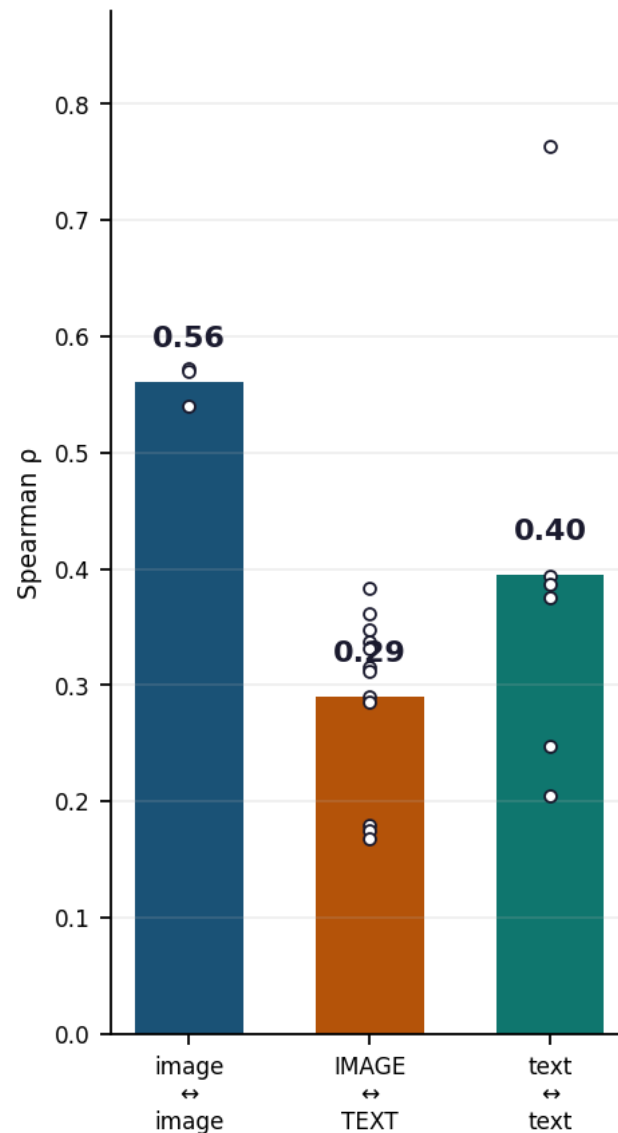
GPT-2 is consistently at the bottom of the pairwise comparisons involving it (roughly 0.168-0.248), consistent with its collapsed geometry carrying less readily usable shape information.

Shape agreement across modalities — seven encoders, four training objectives, no fitted maps

A · All 21 pairs of the seven-space hub measured with NO fitted map



B · Cross-modal agreement is real, and only modestly lower



C · What the cross-modal bar means

The amber group is the surprising one.

A vision encoder that never saw text and a text encoder that never saw an image agree at $\rho \approx 0.31$ on how far apart the same items are — with no map fitted, nothing aligned, and no objective on either side rewarding agreement with the other.

Two comparisons put that in scale:

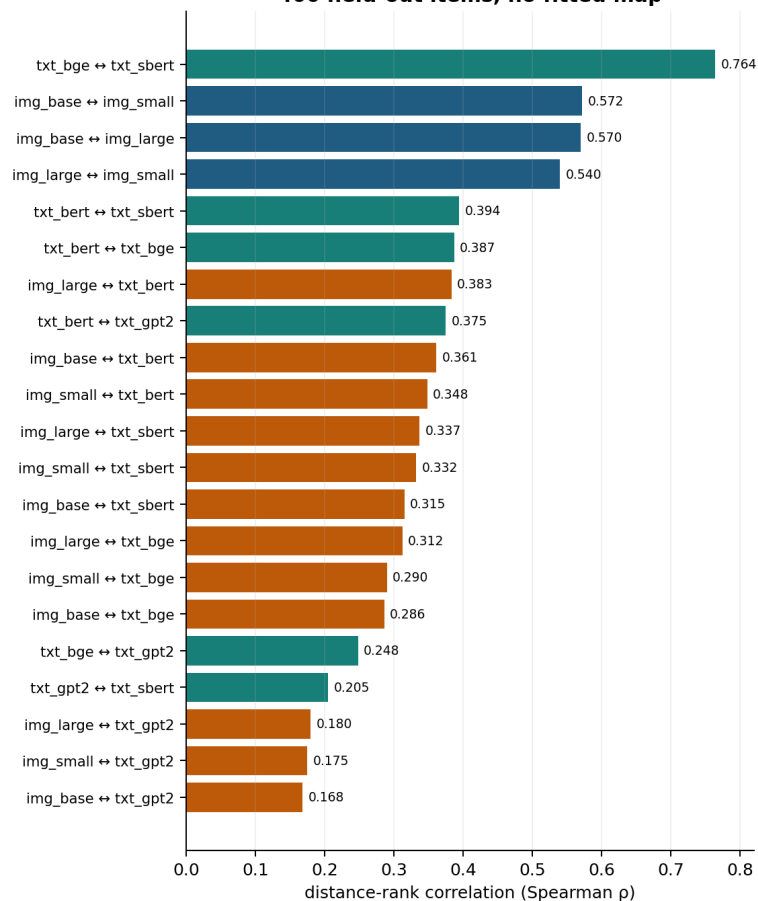
- image ↔ image reaches 0.56 — but those three encoders are distilled from ONE ViT-g teacher, so shared ancestry is doing part of that work.
- bge ↔ SBERT reaches 0.76, the highest of all, and they share no lineage — only an OBJECTIVE. Shared training goal beats shared ancestry here.

GPT-2 sits at the bottom of every group (0.17–0.25): its collapsed space encodes less of the shared structure, not merely less readably.

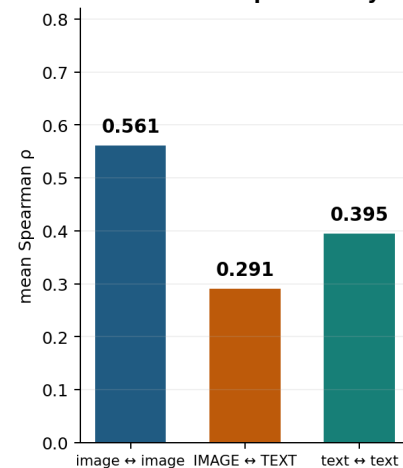
G1 updated: shape agreement, shared space, and geometry - 7 encoders

DINOv2 small/base/large + bge-m3 + GPT-2 + BERT + SBERT | 9,533 common items

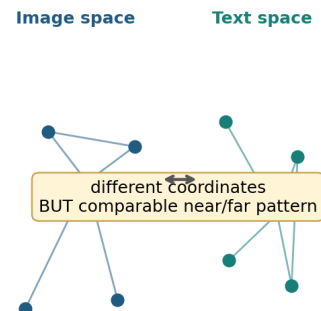
Test 1 - Pairwise shape agreement 400 held-out items, no fitted map



Test 2 - Group summary



What “shared geometry” means



How to read the tests

Similarity

Distance-rank agreement. High ρ = similar relative geometry; no map is fitted.

Shared space

One 512-d Hub. Each encoder learns its own map into the same coordinates.

Transport

Direct A→B vs A→Hub→B. R@1 retention measures usable correspondence after composition.

Geometry

Collapsed targets can be readable as sources but difficult to rank/write into (GPT-2 is the clearest case).

Key finding: cross-modal agreement exists before fitting (mean ρ = 0.291), so the Hub supplies common coordinates for a correspondence that is already partly present.

23. Test 2 - Build one balanced 512-d Hub from all seven spaces

The seven standardized blocks are concatenated per aligned item:

$$768 + 1536 + 2048 + 1024 + 768 + 768 + 768 = 7680 \text{ dimensions}$$

Using the 8,533 training rows, a balanced whitened-PCA basis is fitted and truncated to 512 dimensions:

$$Z_{\text{train}} \text{ in } R^{(8533 \times 7680)} \rightarrow \text{PCA/SVD} \rightarrow \text{whiten} \rightarrow H_{\text{train}} \text{ in } R^{(8533 \times 512)}$$

Question: can seven encoders with different widths, two modalities, and four text objectives share one coordinate system?

Each original space then gets one ridge map W_k that predicts the same Hub coordinates H . There are N encoders and N maps - not N -squared pairwise adapters.

Held-out ability to reach the Hub

Space	Held-out Hub R2
img_small	0.367
img_base	0.533
img_large	0.591
txt_bge	0.288
txt_gpt2	0.228
txt_bert	0.211
txt_sbert	0.265

A low value here is a warning that reconstruction into or through that space may be difficult, but it is not by itself the final portability verdict; collapsed target geometry can make R2 especially misleading.

24. Test 3 - Direct A-to-B map versus A-to-Hub-to-B composition

Question: if a direct pairwise map $A \rightarrow B$ is the reference, how much usable retrieval correspondence survives when the same transfer is forced through the shared Hub?

$$\text{direct: } A \rightarrow B \quad \text{versus} \quad \text{composed: } A \rightarrow H \rightarrow B$$

Headline metric: use scale-free $R@1$ retention rather than mean R2 retention whenever a target has low effective rank. Across all 41 scorable ordered pairs, mean

R@1 retention is **87.0%**. Excluding structurally broken targets, it is **95.6%** over 36 pairs.

Do not use the mean R2-retention subset as the headline. R2 is unreliable for collapsed targets, and the reported R2 mean is computed only over pairs that did not fail, excluding the hardest cases by construction.

Source/target asymmetry

Space	As source	As target	To bge?	Reading
img_small	99.3%	99.6%	112.0%	carried both ways
img_base	94.8%	99.8%	97.6%	carried both ways
img_large	91.1%	98.6%	94.6%	carried both ways
txt_bge	96.3%	98.1%	-	carried both ways
txt_gpt2	95.1%	24.8%	93.9%	READ-ONLY: readable out of, not reliably writable into
txt_bert	99.0%	76.6%	94.2%	carried both ways, weaker as target
txt_sbent	93.8%	101.0%	96.1%	carried both ways

The critical lesson is that a space can be a strong *source* and a weak *target*. GPT-2 is the clearest case: the Hub can read useful information out of GPT-2, but ranking inside GPT-2 is fragile because its own vectors are extremely close to one another. The extra Hub bottleneck error is then larger than the tiny gaps retrieval must preserve.

25. Test 4 - HUB_DIM x alpha transfer sweep

Question: where is the width at which the Hub is wide enough to carry useful structure but not so wide that whitened-PCA tail noise destroys transfer?

HUB_DIM	rows / Hub dim	alpha=.01	alpha=.1	alpha=1	alpha=10
128	66.7	0.399	0.399	0.401	0.401
256	33.3	0.427	0.427	0.428	0.427
512	16.7	0.451	0.450	0.449	0.448
768	11.1	0.006	0.018	0.078	0.311
1024	8.3	0.005	0.005	0.039	0.135

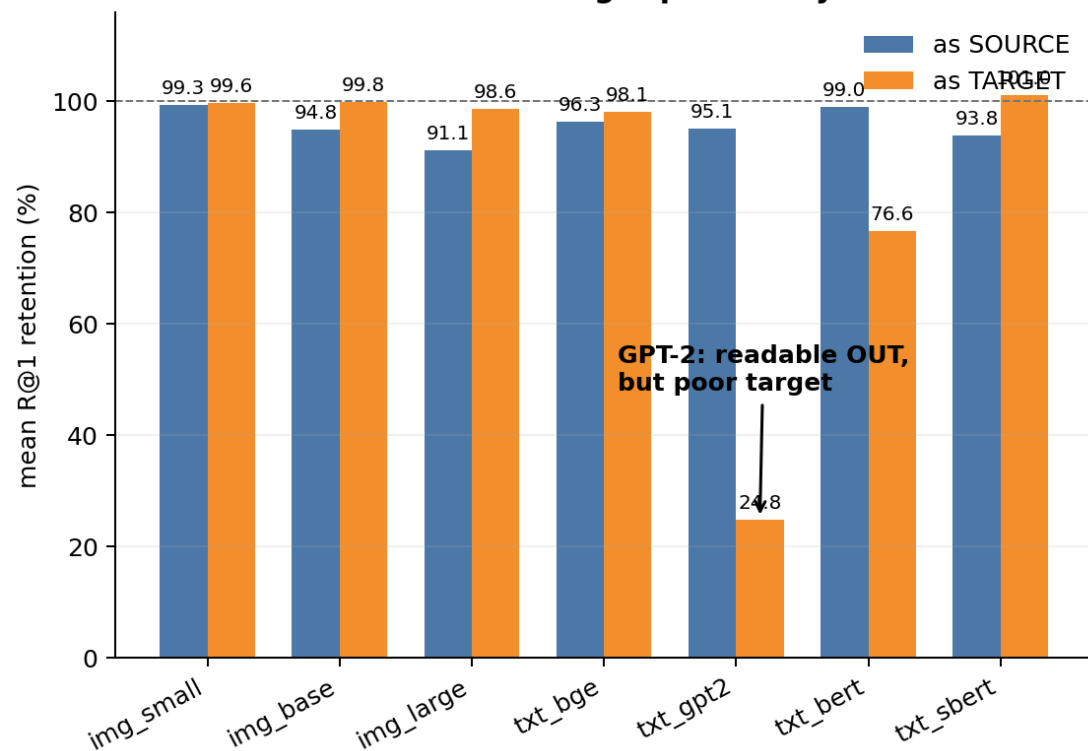
Best transfer: HUB_DIM = 512, alpha = 0.01, R@1 = 0.451, verification AUC = 0.986.

The sharp cliff after 512 is not explained by too few rows per Hub dimension: even 1,024 dimensions has 8.3 rows per Hub dimension. The measured mechanism is

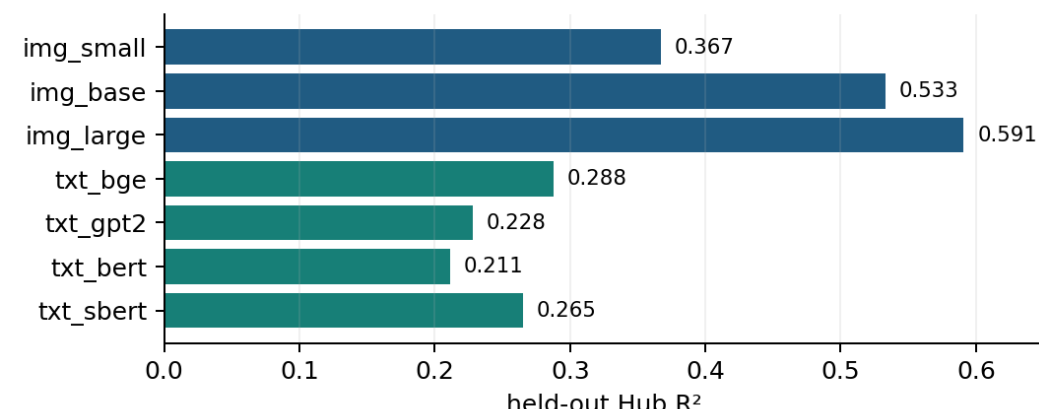
whitened-PCA noise amplification in low-variance tail directions. Alpha can soften the cliff, but cannot replace the width choice.

G1 updated: Hub portability, target geometry, and the 512-d transfer optimum

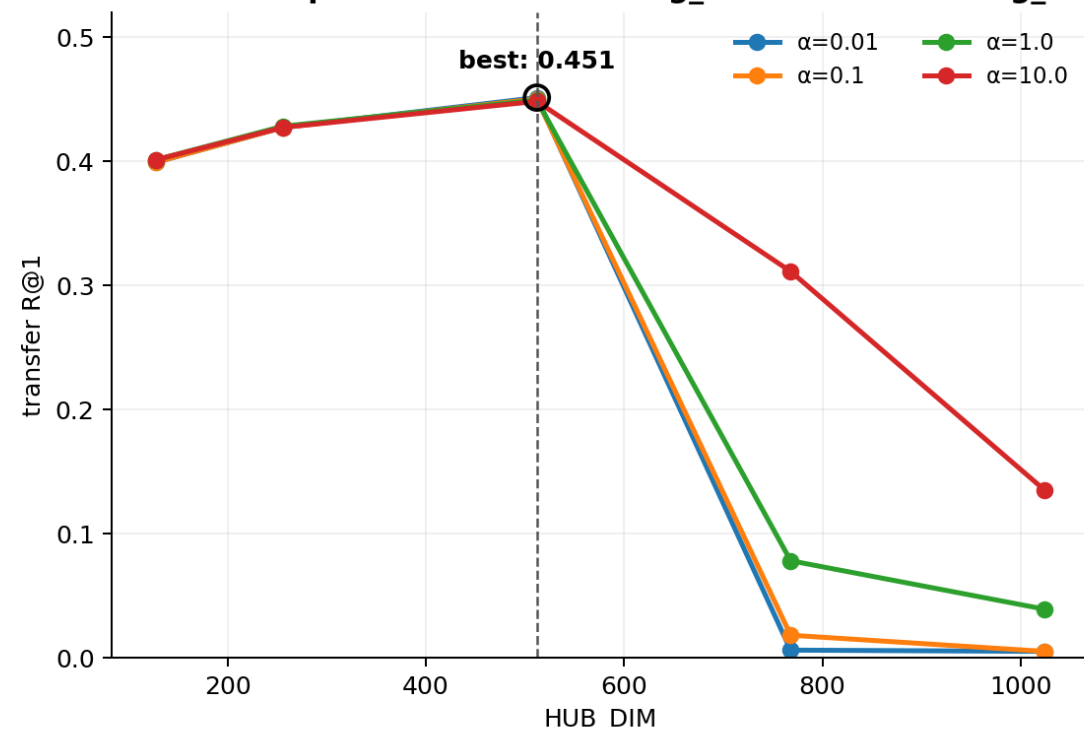
Source vs target portability



How well each original space reaches the 512-d Hub



Transfer sweep: head trained on img_small → unseen img_base



Scale-free verdict and interpretation

87.0% mean R@1 retention over all 41 scorable ordered pairs

95.6% mean R@1 retention over 36 pairs excluding structurally broken targets

512 best Hub width for transfer; wider whitened-PCA tails amplify noise

0.986 verification AUC at the best 512-d transfer setting

Do not use mean R² retention as the headline: collapsed targets make R² unreliable, and failed pairs are excluded from that average by construction.

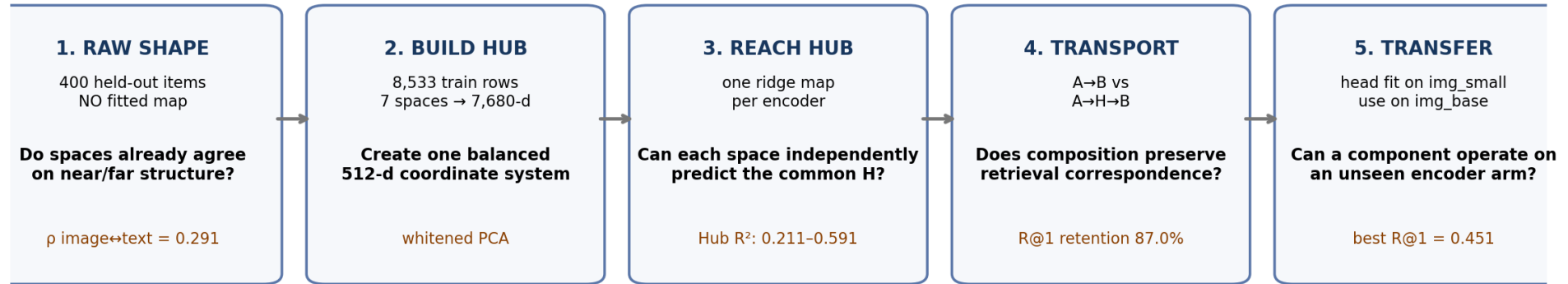
26. Test 5 - What the shared-space visualization is intended to show

The notebook's `plot_hub()` function is a diagnostic with three complementary panels:

1. **Hub projection:** map held-out items from every encoder into H, project the Hub vectors to two dimensions only for display, and connect the seven representations of the same item to their centroid. Tight bundles mean different encoders land the same semantic item near the same Hub location.
2. **Per-space spectrum:** plot singular-value variance shares. A steep spectral cliff means most variance is trapped in very few directions - the geometry is collapsed. Effective rank summarizes this spread.
3. **Same-item vs different-item cosine gap:** compare Hub cosine for the same item across two encoders against cosine for different items. Separation between these distributions is the operational gap that makes cross-encoder retrieval possible.

Conceptual chain: similarity = agreement in relative relationships; shared space = common coordinates; geometry = the near/far and spectral structure that determines whether those coordinates are usable for ranking.

G1 test flow - what each measurement asks



SIMILARITY

Agreement in relative relationships
(e.g. distance ranks).
It does NOT require same coordinates.

SHARED SPACE

A common coordinate system H.
Each encoder has its own W_k
that maps into the same 512 dimensions.

GEOMETRY

The pattern of near/far relationships
and spectral spread.
It determines whether retrieval is readable.

Core result: correspondence is partly present before fitting; the Hub supplies coordinates; target geometry determines whether the correspondence remains usable for retrieval.

27. Report-ready figure captions

Figure - Shape agreement before fitting

Caption: Pairwise distance-rank agreement among seven representation spaces on 400 held-out aligned items, measured before fitting any map. Image-image pairs average Spearman $\rho = 0.561$, text-text pairs 0.395, and image-text pairs 0.291. The non-zero cross-modal mean shows that part of the correspondence exists in raw geometry before a shared coordinate system is fitted. The strongest pair overall is bge-m3 vs SBERT ($\rho = 0.764$), while GPT-2 lies at the bottom of the groups involving it, consistent with its collapsed geometry.

Figure - Shared-Hub diagnostic

Caption: Visualization of the learned 512-d shared Hub. Panel A projects held-out Hub coordinates to two dimensions only for display and joins the seven encoder representations of each semantic item; tighter bundles indicate stronger coordinate agreement. Panel B shows each original space's variance spectrum and effective rank, exposing collapsed geometries. Panel C compares same-item and different-item cross-encoder cosine distributions in the Hub; their separation is the gap that supports retrieval.

Figure - Transfer sweep

Caption: Zero-shot cross-scale head transfer as a function of Hub width and ridge alpha. A caption head fitted on img_small Hub vectors is applied unchanged to unseen img_base Hub vectors. Transfer improves from 128 to 512 dimensions, peaking at $R@1 = 0.451$ ($\alpha = 0.01$; verification AUC = 0.986), then collapses sharply at 768 and 1,024 dimensions. The cliff persists despite adequate rows per Hub dimension, supporting the interpretation that whitened-PCA tail directions amplify noise and dominate cosine geometry beyond the transfer-optimal width.

Figure - Source/target asymmetry

Caption: Mean $R@1$ retention separated by whether each representation space is used as a source or as a target. Most spaces are carried well in both directions, but GPT-2 is strongly asymmetric: 95.1% as a source versus 24.8% as a target. This identifies GPT-2 as effectively read-only at this Hub width and shows that failure to write into a space can arise from the target's own collapsed ranking geometry rather than from loss of information in the Hub.

28. One-slide explanation

1 - Raw geometry: before any fitting, cross-modal distance ranks already correlate at $\rho = 0.291$.

2 - Shared coordinates: seven standardized spaces are combined into a balanced 512-d whitened-PCA Hub; one ridge map per encoder predicts the same H.

3 - Portability: direct A->B is compared with A->H->B. Mean R@1 retention is 87.0% over all 41 scorable pairs and 95.6% after excluding structurally broken targets.

4 - Geometry matters: GPT-2 is readable as a source (95.1%) but not reliably writable as a target (24.8%), because its items are packed too closely for stable ranking.

5 - Width has an optimum: transfer peaks at 512 dimensions (R@1 0.451, AUC 0.986) and collapses beyond it as whitening amplifies low-variance tail noise.

Headline: the correspondence is partly present before fitting; the Hub supplies common coordinates; target geometry determines whether the correspondence remains usable for retrieval.